ME2201: Kinematics and Dynamics of Machinery

Tutorial 4

Submission Date: 28-Aug-2026, in class

1. The Whitworth quick return mechanism is shown in Figure 2, and the link numbers are displayed in the figure. Use the following link lengths: O_2O_4 = 1 cm, O_2A = 2 cm, O_4B = 2 cm, and BC = 4 cm. Determine the angular velocities of all the links using 1) Instantaneous Center Method, and 2) Velocity Polygon Method, when link 2 is making an angle of 90° with link 1 as shown in the figure and is rotating at a constant angular velocity of 1 rad/s in the anticlockwise direction.

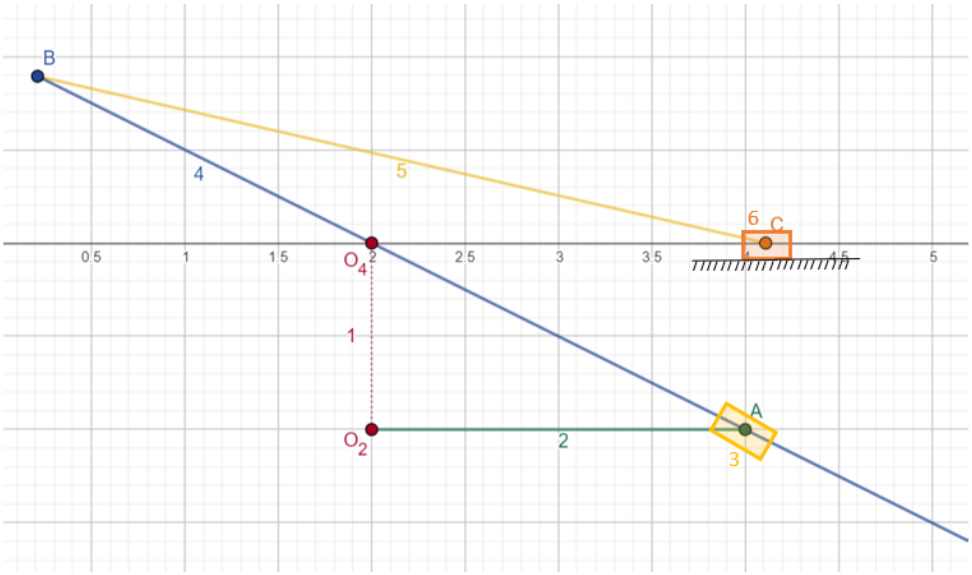


Figure 1: Whitworth mechanism

2. The kinematic diagram of a 6-link mechanism is shown in Figure 2 with link lengths indicated as shown. Link O_2A rotates at a constant angular velocity of 10 rad/s in the counterclockwise direction. Using the instantaneous centers method, determine the angular velocities of links AB and CD. Indicate the direction of rotation of AB and CD on the figure (clockwise or counterclockwise). Note that O_4BC is a rigid link.

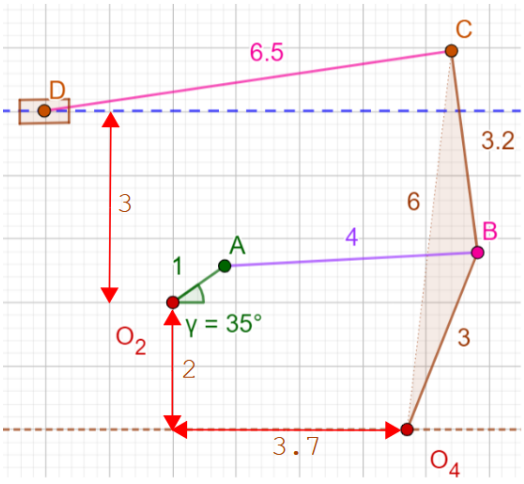


Figure 2

3. (Optional) Figure 3 shows the mechanism used in a two-cylinder 60° V-engine with an articulated connecting rod. Crank 2 rotates in the clockwise direction at a speed of 2000 rpm. Determine the velocities of sliders at B and D, when the link O_2A is horizontal. Given, O_2A = 2 cm, AC = 2 cm, AB = BC = 6 cm, CD = 5 cm.
    a. Using the instantaneous centers method (Locate the following: I_{24}, I_{13}, I_{15}, I_{25}, I_{26}).
    b. Using the velocity polygon method.

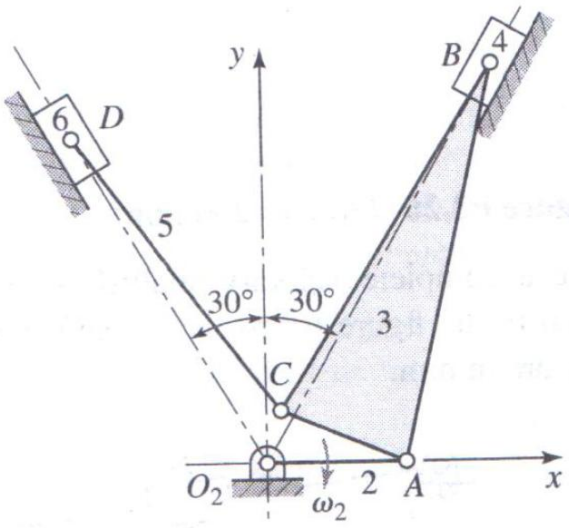


Figure 3